

What is AI?

Thinking and Acting
Humanly and Rationally

Acting Humanly : Turing Test

Turing Test is for testing if the computer can trick a person interrogator to believe that it is a human.

To satisfy above requirement a computer will need these capabilities.

- NLP to communicate successfully.
- Knowledge Representation to store what it knows
- Automated Reasoning to use stored information to answer questions.
- ML to adapt to new circumstances and detect patterns.

Total Turing Test the computer would also need

- Computer Vision to perceive objects
- Robotics to manipulate objects

Thinking Humanly : Cognitive Modelling Approach.

Using logical test against the AI. to see if it's able to pass.

GPS or General Problem Solver was invented.

Acting Rationally : Laws of thought Approach

By using logic invented by Aristotle to solve problems,
However, it is difficult in practice.

Thinking Rationally: Rational Agent Approach.

Agent - Something That Acts

Rational Agent - Something That Acts Rationally.

This approach adapts the acting humanly approach.

because that would allow Thinking Rationally possible.

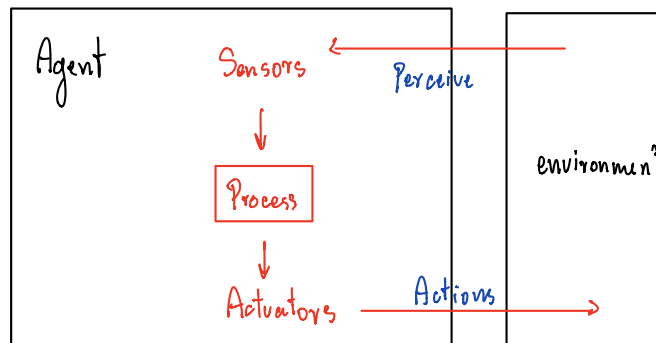
Agent and Environment

Agent - something that acts

Agent could → Perceive the environment using **SENSORS**

→ Process the perceived environment.

→ Acts accordingly, using **ACTUATORS**



Performance Measure

Measure what you want in the environment, rather than what you want it to work

Task Environment - PEAS for an AGENT

P - Performance Measure

E - Environment

A - Actuators

S - Sensors

Properties of the task environment

Observable Fully Observable $\rightarrow$ complete environment state for each point of time
Partially Observable $\rightarrow$ partially, vice versa.

Agents Single Agent $\rightarrow$ Only One agent in the environment
Multi Agent $\rightarrow$ Might be competitive such as CHESS
or Cooperative such as autonomous taxi.

Deterministic Deterministic $\rightarrow$ next environment state is determined completely
Stochastic $\rightarrow$ otherwise

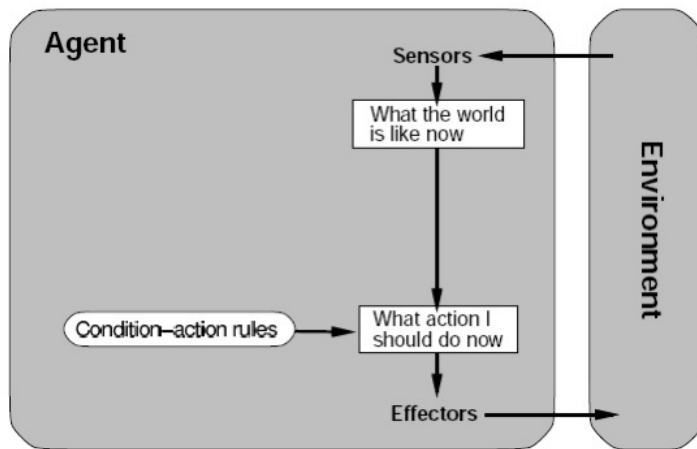
Episodic Episodic $\rightarrow$ Agent's experience is atomic, the action doesn't depend
On the previous one
Sequential $\rightarrow$ Agent's action depends on the previous one and it would affect next one.

Static Static $\rightarrow$ World does not change.
Dynamic $\rightarrow$ World Changes through passage of time.

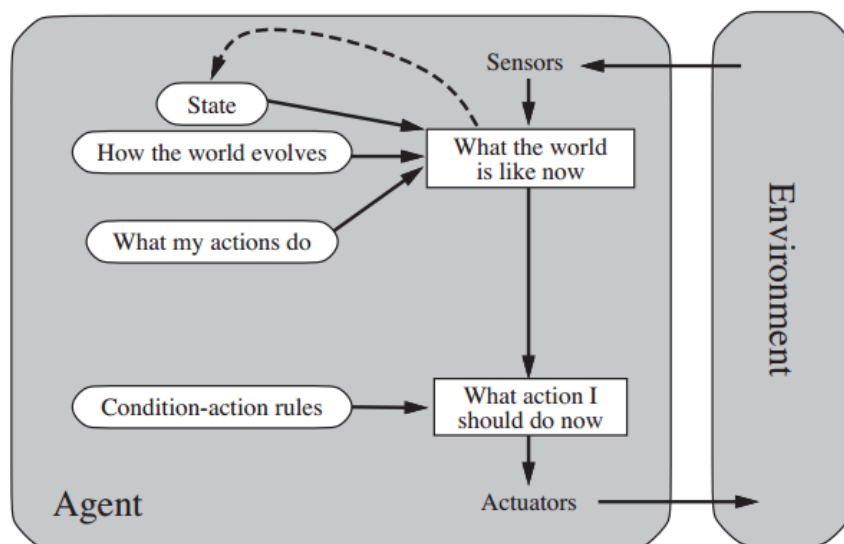
Discrete Discrete $\rightarrow$ State of world over time, limited such as in chess world
Continuous $\rightarrow$ not as smooth overtime, such as taxi driver.

Agent Programs

take the same definition of Agent Concept

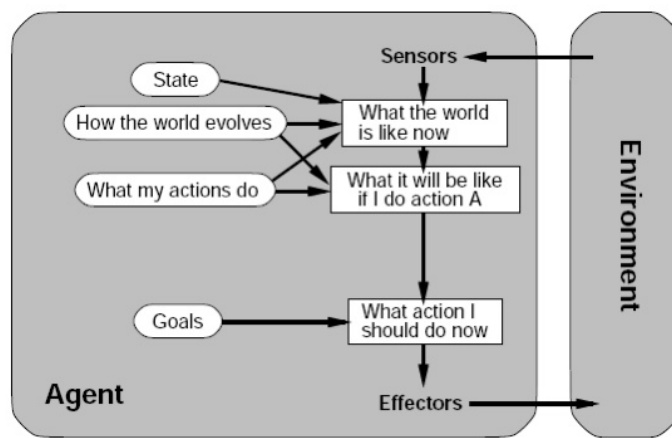


Simple - Reflex - Agent → condition - action rule
also. responds directly to percepts.



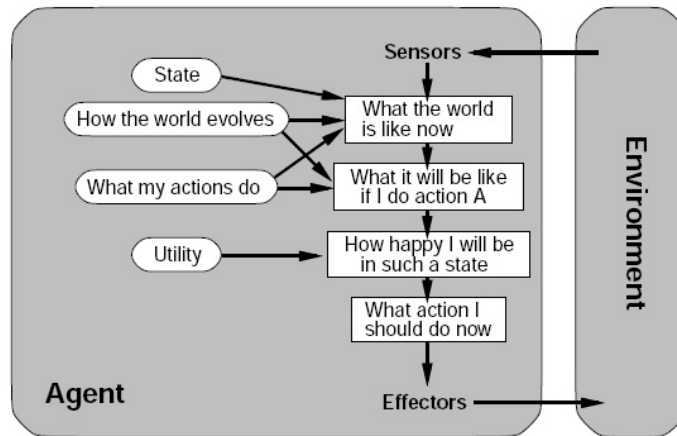
Model-Based reflex agents

keep track of the world it can't see now by having some internal state and update its state before search for action using rules



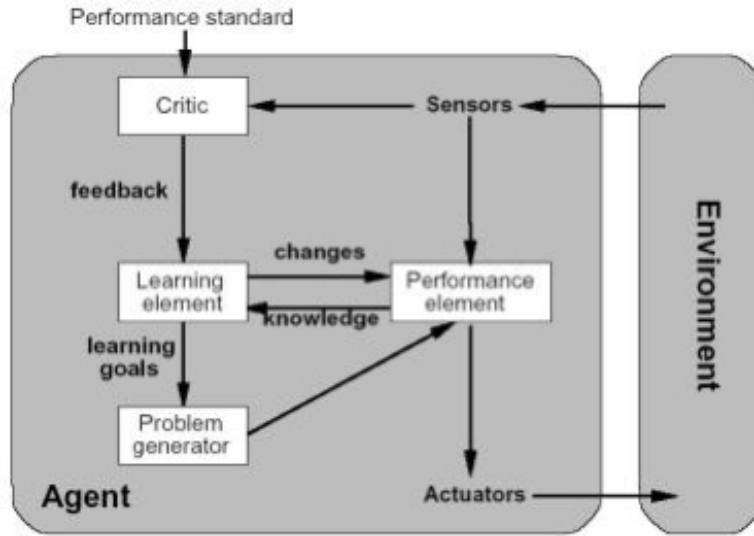
Goal-Based agents

acts to achieve their goals



Utility-Based agents

acts to maximize its own "happiness"



Learning Agents

can be applied to any agent to automatically improve its performance.